

Roman Marach

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EDUCATION

New Jersey Institute of Technology

Bachelor of Science in Computer Science

Newark, NJ

Expected - July 2026

Projects

University Loan Decoder (Claude Hackathon Project) | Python, Gemini API, OCR Feb 2026 – May 2026

- Built an AI-powered financial aid companion that helps university students understand award letters, tuition bills, scholarships, and loan documents in plain English.
- Developed a document-processing pipeline that extracts text from uploaded PDFs/images and sends the content to Gemini API for structured financial analysis.
- Designed strict JSON outputs to populate dashboard cards, red flags, action items, financial terms, and personalized explanations.

AI Document Scanner Tool | Python, PaddleOCR, OCR Pipelines Sep 2025 – Dec 2025

- Engineered a Python-based OCR tool to process scanned pages and extract handwritten and printed text from documents.
- Structured the project as a reusable pipeline for document parsing and text extraction across multiple file types.
- Tested and refined the system against messy real-world inputs, including low-quality scans and handwritten content.

Neural-Network Model | Python, Matplotlib/Seaborn, Scikit-learn, TensorFlow June 2024 – Dec 2024

Created LSTM model designed to predict stock prices based on historical data. The model architecture includes two stacked LSTM layers, followed by dense layers to process and output a single prediction. The model is trained using the Adam optimizer and mean squared error as the loss function, with predictions rescaled back to the original price range for interpretation. Adjustments, such as residual bias correction, are applied to improve accuracy, especially for near-term forecasting.

EXPERIENCE

Software Developer Intern June – August 2024

Wakefern Food Corp

Edison, NJ

- Designed and implemented a program using JavaScript and Node.js to securely and efficiently store online FTP files, ensuring they remained readily accessible for employee use.
- Participated in agile/scrum meetings, contributing insights and ensuring team alignment with project goals.
- Leveraged problem-solving skills to address software issues, collaborating with cross-functional teams including information technology and operations teams.

Help Desk Specialist Feb. 2021 – 2022

Wakefern Food Corp

Edison, NJ

- Thoroughly documented hardware issues, repairs, and upgrades, providing valuable insights for improving system reliability and proactive resolution.
- Troubleshooting and maintaining enterprise hardware systems to ensure operational performance and security.
- Effectively communicated complex technical concepts to non-technical employees, demonstrating strong verbal communication skills and conflict resolution capabilities.

Information Technology Analyst Mar. 2017 – July 2021

Rapid Auto Group

Newark, NJ

- Led a company-wide migration from a paper-based system to a digital platform, optimizing data management and improving information systems workflows by reducing paperwork by 45%.
- Proficiently managed the installation and configuration of workstations across the entire organization, tailoring RAM, processing power and storage allocations to optimize performance based on the specific workloads and requirements of individual employees.

TECHNICAL SKILLS

Languages/Frameworks: Java, Python, C++, SQL, JavaScript, HTML/CSS, Robot, Pandas, NumPy, Bootstrap